

Elliptic Curves in Lean

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Bridging Lean and the LMFDB

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Formalization can be the motivation for discovery of new mathematics.

Definition and group structure

Definition

- An elliptic curve over a field K is defined as the combination of W : `WeierstrassCurve K` and `W.IsElliptic`.
- A `WeierstrassCurve K` is simply a 5-tuple $(a_1, a_2, a_3, a_4, a_6)$ of elements in K (Buzzard & Cremona 2020), which defines the “Weierstrass curve” with equation

$$Y(Y + a_1X + a_3) = X^3 + a_2X^2 + a_4X + a_6.$$

- `IsElliptic` is simply the condition that the discriminant, an explicit polynomial in the five coefficient, does not vanish. It is equivalent to the smoothness of the curve over K .
- The abstract definition (smooth proper curve of genus 1) has been stated during AIM workshop 2024 by Andrew Yang, but connection to this concrete definition hasn't been made.

Rational points

Mathlib has API for rational points of an elliptic curve in affine, projective, and Jacobian coordinates:

- affine: $\{(x, y) \mid W(x, y) = 0\} \cup \{O\}$
- projective: $\{(x, y, z) \mid W^P(x, y, z) = 0\} / (x, y, z) \sim (ux, uy, uz)$
- Jacobian: $\{(x, y, z) \mid W^J(x, y, z) = 0\} / (x, y, z) \sim (u^2x, u^3y, uz)$

The affine version (`Affine.Point`) is the default, since it's the most explicit (doesn't involve quotients).

The actual definitions contain nonsingularity conditions, e.g. $W_X(x, y) \neq 0 \vee W_Y(x, y) \neq 0$, but for an elliptic curve these are automatic.

Rational points

The API for the three coordinate systems are enclosed under the `Affine`, `Projective` and `Jacobian` sub-namespaces of `WeierstrassCurve` respectively; for example, `Jacobian.Polynomial` is

$$W^J = Y^2 + a_1XYZ + a_3YZ^3 - (X^3 + a_2X^2Z^2 + a_4XZ^4 + a_6Z^6),$$

the (2,3,1)-weighted homogenization of `Affine.Polynomial`,

$$W = Y^2 + a_1XY + a_3Y - (X^3 + a_2X^2 + a_4X + a_6).$$

To talk about points of W in an extension L of the base field K , first base change the curve to L (`Affine.baseChange`, which maps all coefficients to L), then take `Affine.Point` of the curve over L .

There used to be the notation $W(L)$ but it was replaced by $(W/L).Point$ in #36328.

Group structure

Mathlib also defines negative of a rational point and the sum of two rational points as `Affine/Projective/Jacobian.Point.neg/add`. Negation $(x, y) \mapsto (x, -(y + a_1x + a_3))$ is a nontrivial symmetry of the curve in $\text{Gal}(W/\mathbb{P}^1)$ that leaves the double cover projection $x : W \rightarrow \mathbb{P}^1$ invariant, while the sum of two points is the negative of the third point of intersection of the secant line through the two points (or the tangent line if both points are equal) with the curve.

In late 2022, I discovered an elementary proof that addition on the rational points is associative and therefore makes the rational points into a group. David Angdinata and me formalized the proof and published a joint paper on it (*[An Elementary Formal Proof of the Group Law on Weierstrass Elliptic Curves in Any Characteristic](#)*).

Associativity proof

The proof starts by defining the obvious map ϕ from the rational points $W(K)$ to the class group of the coordinate ring (`Affine.CoordinateRing`) $K[W] := K[X, Y]/\langle W \rangle$, sending $(x, y) \in K^2$ to $\langle X - x, Y - y \rangle$ and O to the unit ideal.

For any point $P = (x, y) \in W(K)$, the principal ideal $\langle X - x \rangle \subset K[W]$ factors as $\phi(P)\phi(-P)$, so $\phi(-P)$ is the inverse of $\phi(P)$ in the class group, and ϕ preserves negation.

If P_1, P_2, P_3 are the three points (counting multiplicity) of intersection of a line $L(X, Y) = 0$ with W , then the principal ideal $\langle L \rangle$ factorizes as the product of the three ideals $\phi(P_i)$, so ϕ also preserves addition.

Since the class group is a group, we only need to show that ϕ is injective to show that addition is associative on $W(K)$. Since negation/subtraction is preserved it suffices to show the only preimage of the identity is O , or that $\langle X - x, Y - y \rangle$ can never be principal.

Associativity proof

The ideal $\langle X - x, Y - y \rangle$ has codimension 1 in $K[W]$, while the codimension of a principal ideal $\langle p + qY \rangle$ (with $p, q \in K[X]$) can be computed as the degree of

$$N_{K[W]/K[X]}(p + qY) = p^2 - pq(a_1X + a_3) + q^2(X^3 + a_2X^2 + a_4X + a_6).$$

The three terms have degrees $2 \deg p$, $\leq \deg p + \deg q + 1$, and $2 \deg q + 3$ respectively. The middle term cannot be the largest, so the degree of the norm is $\max\{2 \deg p, 2 \deg q + 3\}$, which can never be 1. □

Shape of Weierstrass polynomial

A similar argument shows that $K[W]$ is always a domain, since a factorization

$$W = (Y + p)(Y + q) = Y^2 + (p + q)Y + pq$$

implies $\deg p + \deg q = 3$ and $\deg(p + q) \leq 1$, which is impossible.

Everything only depends on the shape of the Weierstrass polynomial, specifically that its Newton polygon as a polynomial in Y w.r.t. the degree valuation on $K[X]$ is determined by $(2,0)$ and $(0,-3)$, with $(1,-1)$ lying in the interior.

Consequently everything still works without the `IsElliptic` condition, as long as we understand that $W(K)$ denotes the type of *nonsingular* rational points. $K[W]$ needs not be a Dedekind domain, in which case the class group is defined to consist of classes of invertible fractional ideals only, and for a singular point (x, y) the ideal $\langle X - x, Y - y \rangle$ is not invertible.

Examples

Once we have the `AddCommGroup W.Point` instance, we can write

- `Module.finrank ℤ W.Point` for the rank
- `AddCommGroup.torsion W.Point` for the torsion subgroup
- `AddSubgroup.torsionBy W.Point n` for the n -torsion subgroup.

Sample code:

- [Mazur torsion theorem statement](#)
- Conjectures on ranks, Mordell–Weil, Elkies–Klagsbrun curves in the *Formal Conjectures* repo

Division polynomials and torsion structure

n th multiple formula

Exercise 3.7 of Silverman's GTM 106 aims to develop the theory of division polynomials ψ_n to prove the following formula for the n -th multiple of a point $P = (x, y) \neq O$ on an elliptic curve W :

$$[n]P = \left(\frac{\phi_n(x, y)}{\psi_n(x, y)^2}, \frac{\omega_n(x, y)}{\psi_n(x, y)^3} \right)$$

if $\psi_n(x, y) \neq 0$, and $[n]P = O$ if $\psi_n(x, y) = 0$, where ϕ_n and ω_n are additional polynomials defined in terms of ψ_n . In Jacobian coordinates, the two cases can be combined into one formula

$$[n]P = (\phi_n(x, y) : \omega_n(x, y) : \psi_n(x, y))^J,$$

where $(x : y : z)^J$ denotes the equivalence class of (x, y, z) .

This was formalized by me and David Angdinata in June 2024 (#13782, not merged yet).

Torsion structure

It turns out $\phi_n(x, y)$ and $\psi_n(x, y)^2$ as elements in $K[W]$ are represented by univariate polynomials in $K[X]$, of degree n^2 and $\leq n^2 - 1$ respectively (equality holds if $\text{char } K \nmid n$).

Looking at the x -coordinate $\phi_n(x, y)/\psi_n(x, y)^2$, we see that the degree of the isogeny $[n]$ is n^2 . If $\text{char } K \nmid n$, then $[n]$ is separable and therefore the n -torsion subgroup $W(\overline{K})[n]$ has exactly n^2 elements.

Elementary theory of abelian groups then implies that $W(\overline{K})[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$ (formalized by Peiran Wu) and the Tate module $T_\ell(W)$ is isomorphic to $\mathbb{Z}_\ell^2$ (formalized by Chris Birkbeck using Claude) if $\text{char } K \nmid n, \ell$.

Addition and doubling in Jacobian coordinates

If we have an addition formula

$$(x_1 : y_1 : z_1)^J + (x_2 : y_2 : z_2)^J = (A_X(x_1, y_1, z_1, x_2, y_2, z_2) : A_Y(\dots) : A_Z(\dots))^J$$

and a doubling formula

$$[2](x : y : z)^J = (D_X(x, y, z) : D_Y(x, y, z) : D_Z(x, y, z))^J$$

where A_i, D_i are polynomials in $\mathbb{Z}[a_1, a_2, a_3, a_4, a_6, X, Y, Z]$, then we can recursively define

$$\begin{aligned}(\phi_{2n+1}, \omega_{2n+1}, \psi_{2n+1}) &= (A_X(\phi_n, \omega_n, \psi_n, \phi_{n+1}, \omega_{n+1}, \psi_{n+1}), A_Y(\dots), A_Z(\dots)) \\ (\phi_{2n}, \omega_{2n}, \psi_{2n}) &= (D_X(\phi_n, \omega_n, \psi_n), D_Y(\phi_n, \omega_n, \psi_n), D_Z(\phi_n, \omega_n, \psi_n))\end{aligned}$$

and $[n]P = (\phi_n(x, y) : \omega_n(x, y) : \psi_n(x, y))^J$ will be satisfied.

Andrew Sutherland worked out the A_i, D_i only for short Weierstrass equations in his lecture notes, and we worked out the long version. It is crucial and requires separate verification that the formulas continue to hold when one or both of the summands is O or the sum is.

Reduction to the universal/generic case

If we instead want to prove the $[n]P$ formula for specific pre-defined polynomials ϕ_n, ω_n, ψ_n , we only need to verify the six polynomial identities modulo W , which is the approach suggested in Sutherland's notes. It is natural to work in the **universal coordinate ring** $\mathbb{Z}[a_i, X, Y]/\langle W \rangle$ (a domain of characteristic 0), or its field of fractions, the **universal function field**.

The verification for A_Z and D_Z is very easy and done in the notes, but we've been unable to verify the remaining four identities directly. However, once we know $z_1 = z_2 \neq 0$, then $(x_1/z_1^2, y_1/z_1^3) = (x_2/z_2^2, y_2/z_2^3)$ implies $(x_1, y_1, z_1) = (x_2, y_2, z_2)$, so the verification for A_Z and D_Z reduces us to four identities in the universal function field.

Reduction to the universal/generic case

Since A_i and D_i are addition and doubling formulas, if we write

$$P_n = (x(P_n), y(P_n)) := (\phi_n/\psi_n^2, \omega_n/\psi_n^3),$$

the four identities amount to exactly $P_{2n+1} = P_n + P_{n+1}$ and $P_{2n} = [2]P_n$.

Both identities would follow from $P_n = [n](X, Y)$, so we've reduced the general $[n]P$ formula to the single universal/generic case.

Proof of the universal case

We show $P_n = [n]P$ by verifying the $n = 1, 2$ base cases (involving initial terms $\psi_1 = 1, \psi_2 = W_Y, \psi_3$, and ψ_4) followed by induction. With $W_Y(x, y) = 2y + a_1x + a_3$, ϕ_n and ω_n are uniquely determined by

$$x(P_n) = X - \frac{\psi_{n+1}\psi_{n-1}}{\psi_n^2}, \quad W_Y(P_n) = \frac{\psi_{2n}}{\psi_n^4}$$

from which we immediately get

$$\begin{aligned} x(P_n) - x(P_m) &= \frac{\psi_{m+1}\psi_{m-1}}{\psi_m^2} - \frac{\psi_{n+1}\psi_{n-1}}{\psi_n^2} = \frac{\psi_{m+n}\psi_{m-n}}{\psi_m^2\psi_n^2} \\ x(P_{m-n}) - x(P_{m+n}) &= \frac{\psi_{2m}\psi_{2n}}{\psi_{m+n}^2\psi_{m-n}^2} = \frac{W_Y(P_m)W_Y(P_n)}{(x(P_n) - x(P_m))^2}. \end{aligned}$$

This is compatible with the formula

$$x(P - Q) - x(P + Q) = \frac{W_Y(P)W_Y(Q)}{(x(Q) - x(P))^2},$$

which allows us to deduce that $x(P_{m+n}) = x([m+n](X, Y))$ from $P_i = [i](X, Y)$ for $i = m, n, m - n$.

Proof of the universal case

Similarly, the formula

$$W_Y(P)(x(P) - x(Q)) + W_Y(Q)(x(R) - x(P)) + W_Y(R)(x(P) - x(Q)) = 0$$

for $P + Q + R = O$ reduces us to proving

$$W_Y(P_{k+l}) = \frac{W_Y(P_l)(x(P_k) - x(P_{k+l})) - W_Y(P_k)(x(P_l) - x(P_{k+l}))}{x(P_l) - x(P_k)},$$

which is equivalent to

$$\psi_{2(k+l)}\psi_{k-l}\psi_k\psi_l = (\psi_{2k+l}\psi_{2l}\psi_k - \psi_{k+2l}\psi_{2k}\psi_l)\psi_{k+l}.$$

Being able to use division freely in affine coordinates is really convenient for these computations.

See joint paper [Algebraic proofs for division polynomials](#) with David Anginata for more details.

Elliptic relations

The $x(P_n) - x(P_m)$ formula and the last identity and the $x(P_n) - x(P_m)$ formula are special cases ($E(m, n, 1, 0)$ and $EN(k + l, k, k - l, l)$) of elliptic relations and equivalently elliptic net relations:

$$E(a, b, c, d) : h_{a+b}h_{a-b}h_{c+d}h_{c-d} = h_{a+c}h_{a-c}h_{b+d}h_{b-d} - h_{b+c}h_{b-c}h_{a+d}h_{a-d}$$
$$EN(p, q, r, s) : h_{p+q+s}h_{p-q}h_{r+s}h_r + h_{q+r+s}h_{q-r}h_{p+s}h_p + h_{r+p+s}h_{r-p}h_{q+s}h_q = 0$$

studied in prior work of Ward, Stange, Swart, et al.

Since the division polynomials ψ_{2n+1} and ψ_{2n} are defined recursively using $E(n + 1, n, 1, 0)$ and $E(n + 1, n - 1, 1, 0)$ respectively, it is crucial for us to prove that these 1-parameter families of relations imply all elliptic relations (under the mild condition $h_2h_1 \neq 0$ that is satisfied), which I do in the paper [*On elliptic sequences over commutative rings*](#) (arXiv:2604.05280).

Plücker relations

If we write $p_{ij} = h_{i+j}h_{i-j}$, then $E(a, b, c, d)$ becomes the Plücker relation

$$p_{ab}p_{cd} = p_{ac}p_{bd} - p_{bc}p_{ad}$$

that defines the Plücker embedding of the Grassmannian $\text{Gr}(2, \infty)$.

In my proof that all $E(a, b, c, d)$ follow from the 1-parameter families, I found and used a fancy relation among the Plücker relations

$$\begin{aligned} T(cd|mnrs) &= T(nd|mrsc) - T(rd|mns c) + T(sd|mnrc) \\ &\quad + T(nc|mr sd) - T(rc|mnsd) + T(sc|mnrd) \\ &\quad + T(nr|m s cd) - T(ns|mrcd) + T(rs|mn cd) - 2T(md|nrsc), \end{aligned}$$

where $T(mn|abcd) = p_{mn}E(a, b, c, d)$, together with baby versions like

$$T(cd|mnrc) = T(rc|mn cd) - T(nc|mrcd) + T(mc|nr cd).$$

The RHS terms all end in cd , c or d , allowing us to deduce all $E(a, b, c, d)$ from those with small c and d (e.g. $c = 1$, $d = 0$).

Symmetry among elliptic relations

If you partition the four LHS factors with indices $a + b, a - b, c + d, c - d$ into two groups of two in different ways, it turns out the factors in the RHS terms also do not change and just get partitioned differently. For example, $a + b, c + d \mid a - b, c - d$ leads to $a + c, b + d \mid a - c, b - d$ and $a + d, b + c \mid a - d, b - c$, and shows that $E(a, b, c, d)$ is equivalent to

$$E\left(\frac{a + b + c + d}{2}, \frac{a + b - c - d}{2}, \frac{a + c - b - d}{2}, \left|\frac{a + d - b - c}{2}\right|\right),$$

and similarly the partition $a + b, c - d \mid a - b, c + d$ leads to

$$E\left(\frac{a + b + c - d}{2}, \frac{a + b + d - c}{2}, \frac{a + c + d - b}{2}, \left|\frac{b + c + d - a}{2}\right|\right).$$

Half-integers and D_4 lattice

These transformations may produce half-integers, but of course $E(a, b, c, d)$ still makes sense when a, b, c, d are all half-integers. In this case, $a \pm b$ have opposite parity and so do $c \pm d$; partitioning into two evens and two odds yields equivalence to some integral elliptic relation.

We may think of (a, b, c, d) as lying in the D_4 lattice of quadruples of integers and of half-integers, and $(a + b, a - b, c + d, c - d)$ as lying in the D_4 lattice of quadruples summing up to an even number.

Allowing half-integers is crucial in my inductive proof of all $E(a, b, c, d)$ from the 1-parameter families: $E(6, 2, 1, 0)$ can only be derived from equivalence with $E(9/2, 7/2, 5/2, 3/2)$. In general, if $a > b + 1$, the second transformation in the last page allows us to reduce to a smaller a .

Classification of elliptic sequences over fields

In the paper I also showed elliptic sequences (those satisfying all $E(a, b, c, d)$) over fields come in families of 4, 3, and 2 parameters:

- ① $A \cdot \text{EDS}(B, C, D)^{[1/r]}$ with r odd
- ② $A \cdot \text{ES}_{r,s}(B, D)$ with $r < s \neq 2r$ and r, s of opposite parity
- ③ $A \cdot \text{EDS}(0, C, 0)^{[1/r]}$ with r even

where

- $\text{EDS}(B, C, D)$ is the standard elliptic divisibility sequence of which division polynomials $\psi = \text{EDS}(\psi_2, \psi_3, \psi_4/\psi_2)$ form an example
- $\psi_n^{[1/r]} = \begin{cases} \psi_{n/r} & \text{if } r \mid n \\ 0 & \text{otherwise} \end{cases}$
- $h = \text{ES}_{r,s}(B, D)$ is a certain sequence such that $\psi_n \neq 0$ only when $n \equiv \pm r \pmod{r+s}$.

Scheme of elliptic sequences and questions on its geometry

None of the families contains one another, and each family forms an irreducible subset in the scheme of elliptic sequences,

$$\operatorname{Spec} \mathbb{Z}[h_n]_{n \geq 0} / \langle E(a, b, c, d) \rangle_{a > b > c > d \geq 0}.$$

However, it is unclear whether the closure of some higher dimensional family may contain some lower dimensional family, so it is not known whether the lower dimensional families form irreducible components.

It is also not known whether the scheme of elliptic sequences is reduced.

Integrality of ω_n and translation invariant

The definition of ω_n through

$$2\frac{\omega_n}{\psi_n^3} + a_1\frac{\phi_n}{\psi_n^2} + a_3 = W_Y(P_n) = \frac{\psi_{2n}}{\psi_n^4}, \text{ i.e.}$$

$$\omega_n = (\psi_{2n}/\psi_n - a_1\phi_n\psi_n - a_3\psi_n^3)/2$$

only guarantees that its coefficients lie in $\frac{1}{2}\mathbb{Z}$ (we do know that $\psi_n \mid \psi_{2n}$ since ψ is a divisibility sequence). We instead define

$$\omega_n = \frac{\psi_{n+2}\psi_{n-1}^2 + W_X\psi_{n+1}\psi_n\psi_{n-1}}{\psi_2} + Y\psi_n^3$$

(which appears in Enge's book except for a sign error there) and show it satisfies the defining identity using translation invariance of the expression

$$(h_{n+2}h_{n-1}^2 + h_{n+1}^2h_{n-2} + h_2^2h_n^3)/h_{n+1}h_nh_{n-1}$$

for a standard EDS h .

Bilinear forms and reduction homomorphism

Reduction homomorphism

The affine addition formulas are enough to show that $W(K) \rightarrow W(L)$ is a group homomorphism for any field extension L/K , but not good at showing e.g. the reduction map $W(K) \cong W(R) \rightarrow W(k)$ is a homomorphism, in the setting where R is a valuation ring, K its field of fractions and k its residue field, such that the discriminant Δ_W is nonzero in k .

For this it is better to use division-free formulas, in Jacobian coordinates or projective coordinates. Our Jacobian addition formula $(A_X : A_Y : A_Z)^J$ works away from the diagonal, but we do not know whether the doubling formula $(D_X : D_Y : D_Z)^J$ extends to a neighborhood of the diagonal.

Fortunately, Bosma and Lenstra (1995) worked out addition laws $(X^{(1)}, Y^{(1)}, Z^{(1)})$ (away from diagonal) and $(X^{(2)}, Y^{(2)}, Z^{(2)})$ (near diagonal), in projective coordinates. All six polynomials are homogeneous quadratic in $\mathbf{P}_1 = (X_1, Y_1, Z_1)$ and in $\mathbf{P}_2 = (X_2, Y_2, Z_2)$.

Simple expressions for Bosma–Lenstra addition laws

This is much better than the affine approach in the proof of Theorem IV.5.3 of Silverman's GTM 151, which divides $W \times W$ into 8 affine charts, but the six polynomials still take 2+ pages to write down.

When performing symbolic computations (in Mathematica) to check compatibility of the addition laws via the cross identities

$$X^{(1)}Y^{(2)} = X^{(2)}Y^{(1)}, \quad X^{(1)}Z^{(2)} = X^{(2)}Z^{(1)}, \quad Y^{(1)}Z^{(2)} = Y^{(2)}Z^{(1)}$$

modulo $W^P(\mathbf{P}_1)$ and $W^P(\mathbf{P}_2)$, I unexpectedly discovered that the polynomials admit very simple descriptions as cross products of 3-tuples of bilinear forms $\mathbf{B}_X$, $\mathbf{B}_Y$ and $\mathbf{B}_Z$: indeed we have (after correcting two errors in $X^{(2)}$ and $Y^{(2)}$):

$$\mathbf{A}^{(1)} := (X^{(1)}, Y_-^{(1)}, Z^{(1)}) = \mathbf{B}_Y \times \mathbf{B}_X$$

$$\mathbf{A}^{(2)} := (X^{(2)}, Y_-^{(2)}, Z^{(2)}) = \mathbf{B}_X \times \mathbf{B}_Z$$

$$\mathbf{A}^{(3)} := (X^{(3)}, Y_-^{(3)}, Z^{(3)}) = \mathbf{B}_Z \times \mathbf{B}_Y,$$

where $(X^{(3)}, Y^{(3)}, Z^{(3)})$ is another addition law written down by Bosma and Lenstra, and $Y_-^{(i)} = -(Y^{(i)} + a_1X^{(i)} + a_3Z^{(i)})$.

The bilinear forms

So the cross products of $\mathbf{B}_X$, $\mathbf{B}_Y$ and $\mathbf{B}_Z$ compute the projective coordinates of the third point of intersection of the line through $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ with W , which means that $\mathbf{B}_X \cdot \mathbf{P} = 0$, $\mathbf{B}_Y \cdot \mathbf{P} = 0$ and $\mathbf{B}_Z \cdot \mathbf{P} = 0$ are three lines through the third point.

In fact, $\mathbf{B}_Y = \mathbf{P}_1 \times \mathbf{P}_2 = (Y_1 Z_2 - Y_2 Z_1, Z_1 X_2 - Z_2 X_1, X_1 Y_2 - X_2 Y_1)$ simply defines the line through P_1 and P_2 , and the others are also not complicated:

$$\begin{aligned}\mathbf{B}_X &= \frac{a_1}{2} \mathbf{B}_Y - \nabla W_s^P \\ \mathbf{B}_Z &= \frac{a_3}{2} \mathbf{B}_Y + \frac{1}{3} \nabla \Psi_s.\end{aligned}$$

(The only coefficients of Ψ not divisible by 3 are the X^3 and Z^3 ones.)

The bilinear forms, explained

Here W^P is the homogeneous cubic Weierstrass polynomial, ∇W^P denotes its gradient, which is a 3-tuple of quadratic forms, and ∇W_s^P is the 3-tuple of associated symmetric bilinear forms in $\mathbf{P}_1$ and $\mathbf{P}_2$, and Ψ is the cubic form

$$\Psi := (|\mathbf{H}| - 2b_2 W^P)/8,$$

where $\mathbf{H}$ is the Hessian matrix of W^P and $|\mathbf{H}|$ its determinant. Ψ also satisfies the relation

$$Z\Psi = Z^4\psi_3(X/Z) + 3XW^P$$

with the third division polynomial

$$\psi_3(X) := 3X^4 + b_2X^3 + 3b_4X^2 + 3b_6X + b_8.$$

$\mathbf{B}_X$ and $\mathbf{B}_Z$ are “nonsymmetric polarizations” that have integer coefficients, of the quadratic forms ∇W^P and $\nabla \Psi$ respectively.

The matrix of bilinear forms

If we form the 3×3 matrix $\mathbf{B}$ whose rows are $\mathbf{B}_X$, $\mathbf{B}_Y$ and $\mathbf{B}_Z$, basic identities for the cross product yields

$$\mathbf{A}^{(2)} \times \mathbf{A}^{(1)} = |\mathbf{B}| \mathbf{B}_X, \quad \mathbf{A}^{(1)} \times \mathbf{A}^{(3)} = |\mathbf{B}| \mathbf{B}_Y, \quad \mathbf{A}^{(3)} \times \mathbf{A}^{(2)} = |\mathbf{B}| \mathbf{B}_Z.$$

Factorizing these cross product is exactly how I originally found the bilinear forms $\mathbf{B}_X$ and $\mathbf{B}_Z$.

Curiously, $\mathbf{B}$ satisfies “skew-symmetry” under transpose, such that $\mathbf{B}^\top(\mathbf{P}_{1-}, \mathbf{P}_2) = \mathbf{B}(\mathbf{P}_1, \mathbf{P}_2)$, where $\mathbf{P}_{1-} = (X_1, -(Y_1 + a_1 X_1 + a_3 Z_1), Z_1)$. This is somewhat mysterious because there is no obvious reason why the coordinate X, Y, Z should be associated to the gradient, the cross product, and the gradient of the Hessian respectively. This is also specific to the Weierstrass equation, while the bilinear forms and most other properties generalize to arbitrary ternary cubic forms.

Vanishing of $|B|$

To show that the addition laws $A^{(i)}$ are compatible, of course we need that $|B|$ vanishes when P_1 and P_2 are on the curve W . This is confirmed by the identity

$$|B| = \Psi(P_2)W^P(P_1) - \Psi(P_1)W^P(P_2),$$

which you may attempt to derive using Euler's identity

$P \cdot \nabla F(P) = 3F(P)$ for cubic form $F = W^P$ or Ψ . However if you carry it out you'll find that you will need the identity

$$\nabla W^P(P_1) \times \nabla \Psi(P_2) + \nabla W^P(P_2) \times \nabla \Psi(P_1) = 2\nabla W_s^P \times \nabla \Psi_s,$$

or equivalently that

$$\nabla W_s^P(P_1, P_2) \times \nabla \Psi_s(P_3, P_4) + \nabla W_s^P(P_3, P_4) \times \nabla \Psi_s(P_1, P_2)$$

is a symmetric quadrilinear form (invariant under S_4). Of course you can check this by brute force, but no good explanation has been found why this should be true, and AIs are only able to produce proofs by reducing to cubics in normal form.

Other properties

Another surprising fact is that if you take the nine bilinear forms in $\mathbf{B}$, expand them in terms of the monomial basis $\{S_1 T_2\}_{S,T=X,Y,Z}$, and take the determinant of the resulting 9×9 matrix, you get exactly the discriminant of the Weierstrass curve, with no constant factor.

Since the formula $\mathbf{A}^{(2)} = \mathbf{B}_X \times \mathbf{B}_Z$ should be a valid doubling formula away from the singular point of W (if one exists), we see that $\mathbf{B}_X(\mathbf{P}, \mathbf{P}) \times \mathbf{B}_Z(\mathbf{P}, \mathbf{P}) = -\frac{1}{3} \nabla W^P(\mathbf{P}) \times \nabla \Psi(\mathbf{P})$ should vanish only when W is singular at P , i.e. $\nabla W^P(\mathbf{P}) = 0$. There are proofs by reduction to normal form found by AI, but no known direct proof.

If you found answers to these questions, please post an answer at <https://mathoverflow.net/questions/510808>.

Derivation of $\mathbf{A}^{(1)}$

With a little effort one sees that

$$\begin{aligned}\mathbf{A}^{(1)} &= \mathbf{B}_Y \times \mathbf{B}_X = (\mathbf{P}_1 \times \mathbf{P}_2) \times \mathbf{B}_X \\ &= (\mathbf{B}_X \cdot \mathbf{P}_1) \mathbf{P}_2 - (\mathbf{B}_X \cdot \mathbf{P}_2) \mathbf{P}_1 \\ &= (\nabla_{\mathbf{P}_1} W(\mathbf{P}_2)) \mathbf{P}_1 - (\nabla_{\mathbf{P}_2} W(\mathbf{P}_1)) \mathbf{P}_2\end{aligned}$$

which is in fact a valid formula to compute the third point of intersection of the line through $\mathbf{P}_1$ and $\mathbf{P}_2$ with a cubic projective hypersurface not necessarily of dimension 1. I haven't been able to locate the formula in the literature. The proof (found by Grok) uses the expansion

$$W^P(s\mathbf{P}_1 + t\mathbf{P}_2) = s^3 W^P(\mathbf{P}_1) + s^2 t \nabla_{\mathbf{P}_2} W^P(\mathbf{P}_1) + s t^2 \nabla_{\mathbf{P}_1} W^P(\mathbf{P}_2) + t^3 W^P(\mathbf{P}_2).$$

Setting $W^P(\mathbf{P}_1) = W^P(\mathbf{P}_2) = 0$ and $s, t \neq 0$, we immediately get $s : t = \nabla_{\mathbf{P}_1} W^P(\mathbf{P}_2) : -\nabla_{\mathbf{P}_2} W^P(\mathbf{P}_1)$.

Notice that given this formula, $|\mathbf{B}| = 0$ simply says that the third point of intersection lies on the line $\mathbf{B}_Z \cdot \mathbf{P} = 0$.

Potential project

These results have not been formalized in Lean and could constitute a project for this workshop. Showing the reduction map is a homomorphism is immediate from what we have, and it will take more efforts to construct the group scheme associated to a Weierstrass equation over a commutative ring.

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Thank You